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DeepFuse: A multi-rater fusion and refinement network for computing silver-standard annotations

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ABSTRACT

Achieving a reliable and accurate biomedical image segmentation is a long-standing problem. In order to train or adapt the segmentation methods or measure their performance, reference segmentation masks are required. Usually gold-standard annotations, i.e. human-origin reference annotations, are used as reference although they are very hard to obtain. The increasing size of the acquired image data, large dimensionality such as 3D or 3D + time, limited human expert time, and annotator variability, typically result in sparsely annotated gold-standard datasets. Reliable silver-standard annotations, i.e. computer-origin reference annotations, are needed to provide dense segmentation annotations by fusing multiple computer-origin segmentation results. The produced dense silver-standard annotations can then be either used as reference annotations directly, or converted into gold-standard ones with much lighter manual curation, which saves experts' time significantly. We propose a novel full-resolution multi-rater fusion convolutional neural network (CNN) architecture for biomedical image segmentation masks, called DeepFuse, which lacks any down-sampling layers. Staying everywhere at the full resolution enables DeepFuse to fully benefit from the enormous feature extraction capabilities of CNNs. DeepFuse outperforms the popular and commonly used fusion methods, STAPLE, SIMPLE and other majority-voting-based approaches with statistical significance on a wide range of benchmark datasets as demonstrated on examples of a challenging task of 2D and 3D cell and cell nuclei instance segmentation for a wide range of microscopy modalities, magnifications, cell shapes and densities. A remarkable feature of the proposed method is that it can apply specialized post-processing to the segmentation masks of each rater separately and recover under-segmented object parts during the refinement phase even if the majority of inputs vote otherwise. Thus, DeepFuse takes a big step towards obtaining fast and reliable computer-origin segmentation annotations for biomedical images.

1. Introduction

The process of obtaining labels of biomedical images has become a common step in image segmentation pipelines of today. In the context of image segmentation, labels refer to segmentation annotations that consist of segmentation masks. Labeling is usually exercised at the beginning of a pipeline when a training dataset is created for some downstream segmentation network to work better, or at the end of the pipeline as a benchmark dataset to understand the quality of the obtained segmentations. The labels in this context are naturally expected to show a trustworthy and accurate content, which is typically obtained and guaranteed by having human experts to do manual annotations of images, or far less often by employing computer programs.

Labeling biomedical images is nevertheless a very demanding task for human annotators. Moreover, it should ideally be performed by domain experts, who are mostly unavailable. Biomedical image datasets thus often end up with sparse sets of human-origin labels (gold-standard corpus), see Table 1 for an example. To overcome the difficulty of obtaining human-origin labels, several medical imaging studies [1,2] have introduced the idea of producing computer-origin labels (silver-standard corpus). These studies have mostly focused on image segmentation where, compared to much simpler detection or classification tasks, automation yields the most significant time savings for human annotators.

The labeling task primarily aims to provide the most reliable labels possible, regardless of the rater or data type. Nevertheless, even the good-quality segmentation labels that are produced by a single annotator suffer from subjectivity and error-proneness of human annotators due to the presence of ambiguous pixels, labeling confusion, and various errors [3–7]. Different annotators have different opinions (so-called inter-annotator variability) and even the same annotator produces different results when repeatedly annotating the same objects (so-called intra-annotator variability). Thus, it would be an ideal scenario to employ several annotators for labeling the same biomedical images to reduce the ambiguity caused by intra- and inter-annotator variability. Note that in this study, annotators refer to human experts, whereas raters refer to both human experts and computer algorithms.

To obtain input segmentation masks in an automated way, numerous segmentation algorithms are available. However, their accuracy is often not even close to the human level for biomedical images. Hence, several

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label fusion algorithms have been proposed [4,5,7–15] to produce more accurate and reliable silver-standard corpora using multiple segmentation sources, typically results of multiple automated segmentation algorithms. It has previously been shown that a collection of relatively weak raters can be fused to form a stronger rater [16,17]. The label fusion algorithms aim to find a consensus of raters. However, there is no consensus on how to get the consensus. Some fusion algorithms use various versions of expectation-maximization strategy [4,8–11,13–15] while others favor modified majority-voting approaches that suppress the influence of relatively worse-performing raters in favor of the other raters [5,7]. Still, the performance of both types of fusion methods is limited since they have no retrieval mechanisms if the majority or all of the inputs fail to segment a particular object region, such as protrusions [12]. In other words, conventional fusion algorithms have very limited capabilities to enlarge any of the given input segmentation masks, which poses a strict limitation on the fusion performance. The SIMPLE [5] and STAPLE [8] fusion methods under-segment cells when a substantial portion of given inputs is under-segmented, leading to a substantial quality gap between conventional fusion methods and human annotators. Recently, deep-learning-based fusion methods have been proposed to reduce such a quality gap [18,19].

A common and practical use case of silver-standard annotations is to be used as a basis for gold-standard ones after curation by a human expert [20]. This technique is intended to be a time-saving solution for creating dense gold-standard annotations. However, when there is a large quality gap between silver- and gold-standard corpora, human experts still spend a long time fine-tuning silver-standard segmentation masks. In order to make silver-standard annotations an efficient basis for gold-standard annotations, fusion methods need to level up. Convolutional Neural Networks (CNNs) have recently been very popular for image analysis tasks [21–25], including instance segmentation [26–28] and multi-modal segmentation [2,20], due to their ability to train specialized filters for given input images. Notice that applying a filter in image processing terminology is the operation of manipulating the value of every pixel in the image in response to the particular content of pixel's spatial neighborhood. A full-resolution CNN-based label fusion method would increase the abilities of a fusion algorithm by providing individual refinement to the input segmentation masks of each annotator, which is not only the correction of local segmentation errors but also filling holes or enlarging of input segmentation masks if necessary. Therefore, it is worth employing CNNs in a multi-rater fusion method to bridge the quality gap between gold-standard and silver-standard segmentation annotations.

In this paper, unlike existing encoder-decoder CNN architectures, we propose a novel full-resolution (because of the absence of any down-sampling layers) CNN-based multi-rater fusion network for biomedical image segmentation masks, called DeepFuse, to compute silver-standard segmentation annotations. Note that DeepFuse is not an ensemble of methods, since it directly takes rater results (segmentation masks) as inputs, and it does not see individual raters. It refines the input segmentation masks based on the learned rater behavior to correct wrongly-segmented pixels, and then fuses them separately for each object to create the final fusion result. This is the reason why we call it a fusion and refinement network. First, we introduce the architecture of DeepFuse and justify the design choices taken. Then, we evaluate the performance of DeepFuse on the hard task of 2D and 3D cell and cell nuclei instance segmentation for a wide range of microscopy modalities, magnifications, cell shapes and densities available from the Cell Tracking Challenge (CTC) [29–31]. In particular, we take all 26 training video sequences for which CTC segmentation silver truth (SEG ST) is already available – it consists of annotations obtained by a majority-voting approach [7], referred to as CTC baseline silver corpora. The DeepFuse performance is evaluated against CTC gold-standard segmentation annotations (SEG GT). Next, we compare its

Table 1
Annotation coverage for the Cell Tracking Challenge (CTC) training datasets with available silver-truth corpora. Total number of all occurrences of all cells (in case if the dataset name includes '-C') or nuclei ('-N') per sequence and over all time points is given by the number of gold-standard tracking markers (# cells TRA GT: Tracking Gold Truth) which are used in this paper as detection markers. Next, there is the number of cells or nuclei with proper human-origin (gold-standard) segmentation annotations (SEG GT: Segmentation Gold Truth) and computer-origin ones -referred to as the CTC baseline- (SEG ST: Segmentation Silver Truth) and their coverages, respectively (right-most three columns). TRA GT consists of simplified detection and tracking markers, which is relatively less time-consuming to produce for human experts than SEG GT is. It is established in the Challenge that every occurrence of every cell or nuclei in the images is labelled with exactly one TRA GT marker. SEG GT is extremely time-consuming to produce for human experts, and hence very low coverage can be observed in most of the sequences. SEG ST relies on the strategy of automatic expansion of TRA GT markers by fusing candidate segmentation masks from several computer algorithms. Hence, the coverage of SEG ST is above 99 % in most of the video sequences.

Dataset	Sequence	# cells TRA GT	# cells SEG GT	# cells SEG ST	%cover. TRA GT	% cover. SEG GT	% cover. SEG ST
BF-C2DL-HSC	01	8675	255	8674	100	2.9	99.9
	02	64503	309	64501	100	0.5	99.9
BF-C2DL-MuSC	01	5411	217	5388	100	4.0	99.6
	02	7295	298	7223	100	4.1	99.0
DIC-C2DH-HeLa	01	1120	98	1104	100	8.75	98.6
	02	1030	97	988	100	9.4	95.9
Fluo-C2DL-MSC	01	428	112	427	100	26.2	99.8
	02	188	116	188	100	61.7	100
Fluo-C3DH-A549	01	30	15	30	100	50.0	100
	02	30	15	30	100	50.0	100
Fluo-C3DL-MDA231	01	364	121	351	100	33.2	96.4
	02	585	103	582	100	17.6	99.5
Fluo-N2DH-GOWT1	01	2058	144	2054	100	7.0	99.8
	02	2517	121	2287	100	4.8	90.9
Fluo-N2DL-HeLa	01	8639	491	8602	100	5.7	99.6
	02	25420	895	25378	100	3.5	99.8
Fluo-N3DH-CHO	01	905	115	905	100	12.7	100
	02	614	117	614	100	19.1	100
Fluo-N3DH-CE	01	23802	102	23785	100	0.4	99.9
	02	22418	102	22317	100	0.5	99.6
Fluo-C3DH-H157	01	240	119	240	100	49.6	100
	02	159	99	159	100	62.3	100
PhC-C2DL-PSC	01	71403	242	71067	100	3.4	99.5
	02	56804	254	56775	100	4.5	99.9
PhC-C2DH-U373	01	765	100	765	100	13.1	100
	02	692	110	692	100	15.9	100

performance with the SIMPLE and STAPLE fusion algorithms, and CTC baseline silver corpora. We also discuss the challenges of DeepFuse and potential improvements to address them. Finally, we qualitatively show that DeepFuse recovers under-segmented object parts based on the knowledge learned from the training data and produces high-quality silver-standard segmentation annotations. On most of the analyzed datasets, DeepFuse produces significantly better segmentation annotations than the chosen competitors and hence reduces the quality gap between silver- and gold-standard annotations. Another important novelty of DeepFuse is that thanks to the specialized refinement filters, DeepFuse can recover under-segmented object pixels or regions (unlike traditional deep-learning-free label fusion methods) even if the majority of raters fail to segment them.

2. Method: DeepFuse architecture

The DeepFuse architecture is an N-to-1 fusion and refinement network powered by CNNs. It can be described as a sophisticated post-processing network that focuses on learning particular patterns in the results of each rater. The proposed network architecture is shown in Fig. 1. It consists of three-stage 5×5 convolutional layers with rectified linear unit (ReLU) activations, followed by a summation layer to aggregate refined inputs and finally a 1×1 convolutional layer with a sigmoid activation layer to produce the final binary segmentation mask that represents at most one object. Each rater is assigned a set of convolutional filters (we also call them refinement layers) to refine their own segmentation masks. The convolutional part of the network refines the inputs, whereas the later parts reduce the dimension of refined inputs and fuse them. The input image size can arbitrarily be chosen. The only restriction is that all N rater results are of the same size. Image resolution is not changed through the network. To match the input and output size, zero-padded convolutions are used. We observed that three-stage convolutional layers achieve better results than the two-stage ones and 5×5 filters achieve better results than 3×3 ones as also observed in the CU-Net architecture [33]. For enhanced learning precision, we train one object at a time. Even if there are multiple objects in a given image, we split them into single-object images in order to improve the learning of detailed object structures and rater behavior, and also to avoid merging of closely-located objects. More specifically, a dataset of K images with a total of L objects is split into L images, each containing one object only. For N raters (with fixed orders), DeepFuse trains N sets of convolutional filters, see Fig. 1. The network produces a single output for a set of N annotations, hence each set of filters is independently trained by the same loss function and updated after every learning epoch. The fusion of 3-D segmentation masks is carried out by a 3-D version of the network that implements a slice-by-slice paradigm. The open source implementation of the network is available at https://github.com/cememreakbas/DeepFuse.

2.1. Training of the network

The Dice Loss function [26,32] is used to train the network. The batch size was set to 5 for all datasets. The models were trained for 100 epochs for all sequences analyzed in this paper. We did not observe any over-fitting; the performance on the training and validation sets did not differ by more than 5 % for all trained sequences. We used sparse gold-standard segmentation annotations (publicly available for all CTC training datasets) for training and validation with a 80:20 ratio. We did not use any data augmentation since at least 100 cell instances were annotated in every examined sequence. We would like to emphasize that the network is trained with the segmentation results of raters; it is not given any other information from individual raters. As mentioned in the ablation studies (Section 3.8), all rater results are trained simultaneously using a single loss function. We observed that simultaneous training outperforms separate training in all available training sequences (Fig. 5). Thus, the final DeepFuse architecture presented in this paper uses simultaneous training approach, i.e., it is an N-to-1 network driven by a single loss function. Furthermore, there is no strict restriction on the input image size. The only limitation is to use the same image size for all N raters through a whole dataset. If one has images of various sizes in a

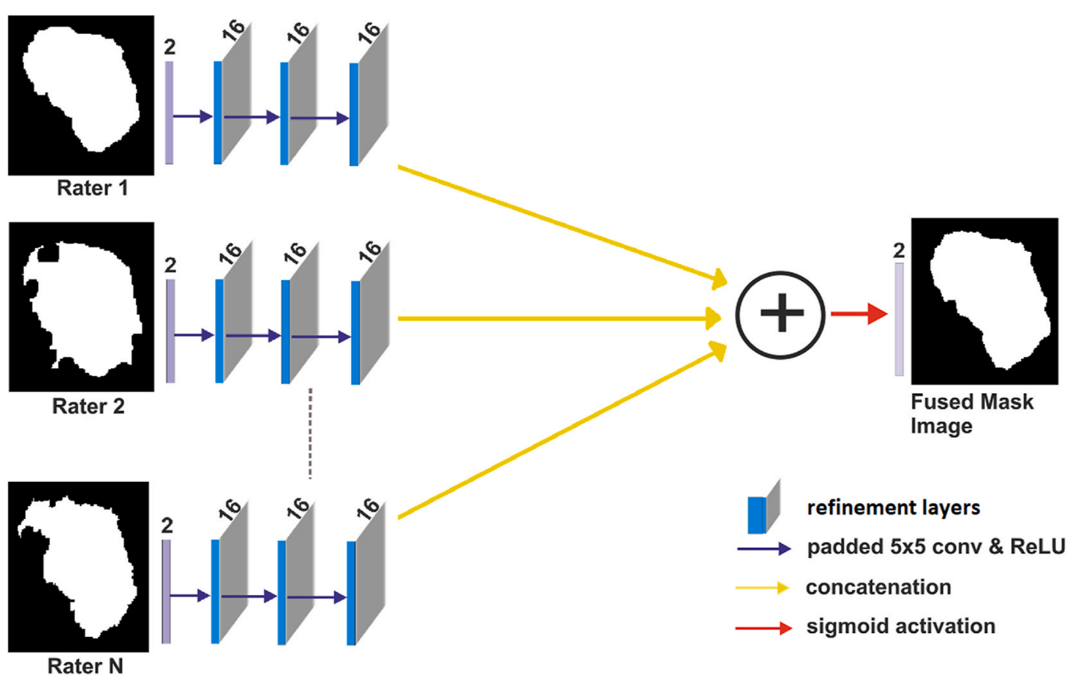

Fig. 1. Illustration of the DeepFuse architecture. For a given biomedical image, such as nucleus or cell, several raters (algorithms) produce their segmentation masks. Each rater is assigned a set of convolutional filters (refinement layers) to refine their own segmentation masks. Then, all those separately refined raters' candidate segmentations are pixel-wise summed and fed to the sigmoid activation to produce the fusion result. Notice that the network includes no down-sampling layers; it preserves the full image resolution, and that the refining layers consists of fixed number of parameters to be learned, allowing the network to process arbitrarily large images. The illustrative image of a cell is taken from the first time-lapse sequence of the PhC-C2DH-U373 dataset.

dataset, zero-padding is the recommended approach to make the dataset ready for training and inference.

3. Results

3.1. Dataset description & input selection criteria

In order to perform an extensive comparison of the fusion algorithms, we used 26 time-lapse microscopy training sequences (16 were 2D + time, 10 were 3D + time) from the Cell Tracking Challenge [29–31]. These sequences show 12 different cell types (Mouse hematopoietic stem cells, Mouse muscle stem cells, HeLa cells used in four sequences, Rat mesenchymal stem cells, GFP-GOWT1 mouse stem cells, Glioblastoma-astrocytoma U373 cells, Pancreatic stem cells, GFP-actin-stained A549 Lung Cancer cells, MDA231 human breast carcinoma cells, *C. Elegans* developing embryo, Chinese Hamster Ovarian (CHO) nuclei, GFP-transfected H157 Lung Cancer cells) acquired using 4 different optical microscopy modalities (Bright-Field, Fluorescence, Differential Image Contrast, Phase Contrast), see Table 2. The image data in these sequences is representative of the variability and challenges encountered in laboratory settings [31].

We used publicly available segmentation algorithms of the Cell Tracking Challenge (CTC) participants as of September 2020 (to have the same inputs as used for the current CTC baseline) and run them on the training sequences of the mentioned datasets to create our inputs, sequences of segmentation masks. The number of used segmentation algorithms varied between 4 and 16 depending on the dataset, by following the same selection criteria as in Ref. [31]. We only used algorithms that reached at least 50 % of the human performance, measured as the average performance on both sequences of a particular dataset and being satisfied on both training and test datasets. We observed that algorithms that performed below this 50 % threshold may contain poorly-segmented masks that may substantially differ from the gold-standard annotations. If there were more than 16 algorithms that satisfied the selection criteria, only the best-16 out of them were used. The gold-standard annotations (dense for detection and sparse for segmentation) are publicly available on the CTC website [29] for all training datasets.

3.2. Experiment setup

For all four compared fusion methods, the strategy was to expand human-origin detection markers (which are relatively cheap to obtain and available for the CTC training datasets) by merging multiple computer-generated segmentation masks. In this paper, we based our study on the publicly available training data from the CTC, which comes with human-origin tracking annotations, from which we essentially neglected the linking part to treat these annotations as detection annotations. Since they are gold-standard detection annotations, they guarantee the absence of false positive (FP) markers, which strengthens the reliability of produced silver-standard annotations.

During the above stage, which is technically a pre-processing stage, we also synchronized label IDs across segmentation sources so that a specific object had the same label ID in all raters. Otherwise, the asynchronous label IDs may be understood and treated as different objects and cause inaccurate fusion results in the SIMPLE and STAPLE fusion processes even though they represent the same object. Next, silver segmentation truth was computed by fusing the input segments, which were here the outputs of a set of automatic segmentation methods. Note that there might be missing objects due to the insufficient segmentation performance of some, sometimes all, of the input sources. In such cases, we did not intervene and fed empty segmentation masks to the methods. The created fused masks constitute the silver-standard segmentation corpus for a given dataset which is, in our case, a time-lapse sequence processed per object in randomized order. Nevertheless, our method can work with a set of static images with detection markers as well.

For each examined sequence, we obtained three sets of silver-standard corpora produced by three tested fusion algorithms (STAPLE, SIMPLE, DeepFuse) with all qualified inputs (the same inputs as CTC baseline -i.e., CTC ST annotations-used). For each fusion method, we gradually increased the quality threshold of segmentation algorithms by decreasing the number of inputs with a step of 2 to explore if using M-best inputs (M = 1, 3, 5, …, N where N is the number of available inputs) can increase the quality of silver-standard segmentation masks. For each sequence-method pair, we chose the best silver annotations in terms of SEG score measured against the corresponding publicly available CTC gold-standard segmentation annotations. We report the winning M numbers and corresponding quality threshold values in Section 3.7 and discuss their impact on the quality of fusion results in Section 4. Finally,

Table 2
Characteristics of the analyzed training datasets. Acronyms: spinning-disk confocal (SDC); laser-scanning confocal (LSC); brightfield (BF); widefield fluorescence (WF); phase contrast (PhC); differential interference contrast (DIC). The numbers in parentheses indicate particular values for the second video in a given dataset in case they differ from the values for its first video.

Name	Objects of interest	Modality/ Magnification /Bit depth	Frame size [pixels]	No. of frames	Spatial resolution [μm]	Temporal resolution [min]
BF-C2DL-HSC	Mouse hematopoietic stem cells	BF/10× /8	1010 × 1010	1764	0.645 × 0.645	5
BF-C2DL-MuSC	Mouse muscle stem cells	BF/10× /8	1070 × 1036	1376	0.645 × 0.645	5
DIC-C2DH-HeLa	HeLa cells	DIC/63× /8	512 × 512	84	0.19 × 0.19	10
Fluo-C2DL-MSC	Rat mesenchymal stem cells	SDC/20× /16	992 × 832 (1200 × 782)	48	0.3 × 0.3 (0.398 × 0.398)	20 (30)
Fluo-C3DH-A549	A549 lung cancer cells	SDC/63× /16	350 × 300 × 29 (400 × 270 × 28)	30	0.126 × 0.126 × 1	2
Fluo-C3DH-H157	H157 lung cancer cells	SDC/63× /16	992 × 832 × 35 (992 × 832 × 80)	60	0.12 × 0.12 × 0.5	1 (2)
Fluo-C3DL-MDA231	MDA231 human breast carcinoma cells	LSC/20× /16	512 × 512 × 30	12	1.24 × 1.24 × 6	80
Fluo-N2DH-GOWT1	Nuclei of GOWT1 mouse stem cells	LSC/63× /8	1024 × 1024	92	0.24 × 0.24	5
Fluo-N2DL-HeLa	Nuclei of HeLa cells	WF/10× /16	1100 × 700	92	0.645 × 0.645	30
Fluo-N3DH-CE	Early *C. elegans* developing embryo	LSC/63× /8	708 × 512 × 35 (712 × 512 × 31)	250	0.09 × 0.09 × 1	1
Fluo-N3DH-CHO	Nuclei of Chinese hamster ovarian cells	LSC/63× /8	512 × 443 × 5	92	0.2 × 0.2 × 1	9.5
PhC-C2DH-U373	Glioblastoma-astrocytoma U373 cells	PhC/20× /8	696 × 520	115	0.65 × 0.65	15
PhC-C2DL-PSC	Pancreatic stem cells	PhC/4× /8	720 × 576	300	1.6 × 1.6	10

we quantitatively compare the performance of the fusion algorithms with the CTC baseline on each sequence.

Regarding the practicality of execution of the fusion algorithms, we had to run the four fusion algorithms on different computers. This fact reflects the time-complexity of the algorithms. In particular, the rough computation times and platforms are as follows:

- SIMPLE: a few minutes (small M) to few hours (large M), on a standard Intel i7 CPU
- STAPLE: a few hours (small M) to few days (large M), even weeks (large M + large images), on a powerful workstation CPU
- DeepFuse: a few hours (small M) to one day (large M), on a powerful NVIDIA A100 GPU (including training times)
- CTC baseline: a few hours to a few years (in terms of CPU hours), on a supercomputer

3.3. Pre-processing of labels

The primary objectives of our pre-processing pipeline are to eliminate false positive segmentation masks (FP) across segmentation sources and to synchronize label IDs across segmentation sources. The former step addresses one of the biggest concerns about computer-origin labels, which is the expectation that an increased abundance of FP objects will mislead the training of supervised algorithms. Our pre-processed labels overcome this concern; they are guaranteed to contain no FP objects. The latter step is needed to pair the same objects across multiple raters for a given object and it is rather straightforward: we simply take the gold-standard detection markers as the reference and synchronize the corresponding input label IDs for the objects that match the detection markers.

In this paper, the label pre-processing is applied as in the CTC baseline fusion method [7], which is described as the expansion of gold-standard tracking markers into full segments (see Fig. 2). In our application, the method essentially iterates over all markers and chooses the corresponding segment from the input instance segmentations for each marker. The criterion for a segment to be chosen is that it covers more than a half of the size of the marker (Fig. 2A). In this way, there can be at most one segment assigned to a marker, but sometimes no segment is chosen at all (Fig. 2C). Not rarely, however, one segment mask satisfies the criterion for multiple markers and is assigned to them. In this case, the pre-processing step attempts to split the mask evenly into multiple masks, one mask per one marker (Fig. 2C). This is achieved with morphological watershed seeded with the markers themselves and constrained by the area of the original large segment (Fig. 2D). Additionally, if multiple connected components exist for a given segment, all except the biggest one are removed. Ultimately, segments take labels from their markers (Fig. 2B and C). To sum it up, during the pre-processing we eliminate the segments that do not cover more than a half of any tracking marker, also all but the biggest connected component, and split over-segmenting segments.

3.4. CTC baseline with combinatorial majority voting (CMV)

CTC baseline silver-standard annotations were produced using the combinatorial majority voting method, which is a combination of two nested algorithms [7]. The inner algorithm computes a fusion output based on fixed-weights pixel-wise majority voting. The outer algorithm visits all non-empty subsets of available segmentation sources in order to find the best possible fusion output. Therefore, in some cases, it is possible that some of the segmentation sources do not contribute to the fusion result at all. The computation cost of the CMV method increases exponentially with the number of input sources. When the number of inputs is larger than 16, the computation time gets too long, even on a supercomputer. Thus, we limited the number of inputs to 16 (lower numbers are recommended on modest hardware). The details of the input selection criteria are described in Section 3.1.

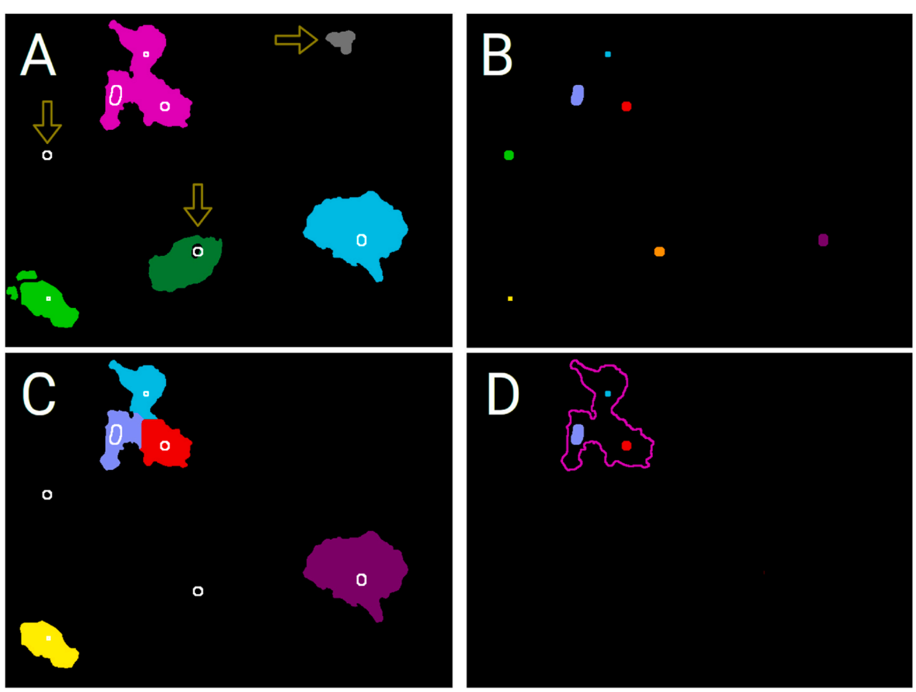

Fig. 2. Illustration of the pre-processing procedure. The input instance segmentation is shown with solid objects (A), input tracking markers are shown as solid objects (B,D) as well as white outlines (A,C). The pairing of segments to markers may fail due to absence of satisfying segments (false negatives, see vertical arrows in A), or presence of extra segments (false positive, see horizontal arrow in A). After the pairing, segments take colors (C) from their markers (B), largest connected components are preserved (bottom left in A and C) and over-segmented segments are split (top in C) with morphological watershed seeded from the markers and constrained with the original large segment (D). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Table 3
Quantitative comparison of fusion algorithm performance (1st subcolumn: Their segmentation accuracy, 2nd subcolumn: Winning M values indicating the optimal number of inputs – see Section 3.2, 3rd subcolumn (for DeepFuse only): Input quality threshold that corresponds to the best M^{th} rater's segmentation accuracy), best individual input raters (may differ for each sequence), and the human annotators (i.e., inter-annotator variability). The CTC SEG measure [29–31] was used, which measures average Jaccard similarity to the CTC human-origin segmentation annotations. Bold numbers indicate the best fusion result among four fusion methods for each sequence. The symbols *, + and • indicate that for the given sequence, the results of DeepFuse are statistically significantly better than SIMPLE, STAPLE and CTC Baseline, respectively, with a p-value< 0.05. Please note that quality thresholds observed below 50 % are still valid because 50 % criterion is applied on the average performance on both sequences of a particular dataset (See Fluo-C3DH-H157 row for an example).

Dataset	Seq.	SEG Score [0, 1] (1: the best)										# Raters
		SIMPLE/Winning M		STAPLE/Winning M		CTC Baseline	DeepFuse/Winning M/Corresp. Quality Threshold			Best Individual Input Rater	Human Perf.	
BF-C2DL-HSC	01	0.898	1	0.895	1	0.898	**0.915**$^{*+\bullet}$	5	64.6 %	0.895	0.892 ± 0.036	5
	02	0.853	3	0.853	1	0.853	**0.879**$^{*+\bullet}$	5	63.6 %	0.847		
BF-C2DL-MuSC	01	0.820	3	0.823	3	0.833	**0.837**$^{*+}$	3	65.6 %	0.814	0.843 ± 0.026	4
	02	0.784	1	0.784	1	0.784	**0.796**$^{*+\bullet}$	4	44.6 %	0.776		
DIC-C2DH-HeLa	01	0.961	1	0.961	1	0.965	**0.974**$^{*+\bullet}$	11	48.8 %	0.961	0.784 ± 0.066	11
	02	0.961	1	0.961	1	0.965	**0.972**$^{*+\bullet}$	11	54.5 %	0.952		
Fluo-C2DL-MSC	01	0.779	1	0.766	1	0.792	**0.805**$^{*+\bullet}$	7	49.3 %	0.729	0.769 ± 0.047	11
	02	0.762	3	0.746	11	0.817	**0.846**$^{*+\bullet}$	11	47.6 %	0.762		
Fluo-C3DH-A549	01	0.975	1	**0.991**	1	**0.991**	0.989*	5	89.9 %	0.991	0.859 ± 0.020	7
	02	0.954	1	**0.991**	1	**0.991**	0.989*	1	95.8 %	0.991		
Fluo-C3DL-MDA231	01	0.729	5	0.718	5	**0.739**	0.727^{+}	7	45.8 %	0.684	0.742 ± 0.048	7
	02	0.734	5	0.738	5	**0.752**	0.730	3	67.5 %	0.705		
Fluo-N2DH-GOWT1	01	0.913	1	0.920	16	0.933	**0.943**$^{*+\bullet}$	13	72.2 %	0.913	0.886 ± 0.062	16
	02	0.976	1	0.976	1	0.976	**0.982**$^{*+\bullet}$	9	89.3 %	0.954		
Fluo-N2DL-HeLa	01	0.874	3	0.872	13	0.884	**0.889**$^{*+}$	15	68.2 %	0.860	0.904 ± 0.035	16
	02	0.901	3	0.898	11	0.909	**0.914**$^{*+\bullet}$	11	81.4 %	0.891		
Fluo-N3DH-CHO	01	**0.967**	1	0.853	11	0.883	0.883^{*+}	5	80.0 %	0.852	0.904 ± 0.081	15
	02	**0.933**	15	0.928	13	0.931	0.931	5	82.6 %	0.920		
Fluo-N3DH-CE	01	N/A	N/A	N/A	N/A	0.705	**0.709**	5	62.4 %	0.745	0.844 ± 0.021	8
	02	N/A	N/A	N/A	N/A	0.711	**0.714**	5	56.8 %	0.725		
Fluo-C3DH-H157	01	0.845	5	0.891	1	0.901	**0.904**$^{*+}$	6	60.1 %	0.891	0.924 ± 0.009	6
	02	0.799	6	0.779	1	0.824	**0.841**$^{*+\bullet}$	6	49.8 %	0.779		
PhC-C2DL-PSC	01	0.779	9	0.779	9	0.785	**0.796**$^{*+\bullet}$	11	61.5 %	0.764	0.788 ± 0.044	16
	02	0.780	3	0.780	3	0.784	**0.800**$^{*+\bullet}$	13	59.6 %	0.779		
PhC-C2DH-U373	01	0.949	3	0.948	9	0.951	**0.957**$^{*+\bullet}$	13	81.4 %	0.947	0.836 ± 0.143	15
	02	0.884	1	0.882	9	0.888	**0.898**$^{*+\bullet}$	7	80.7 %	0.880		

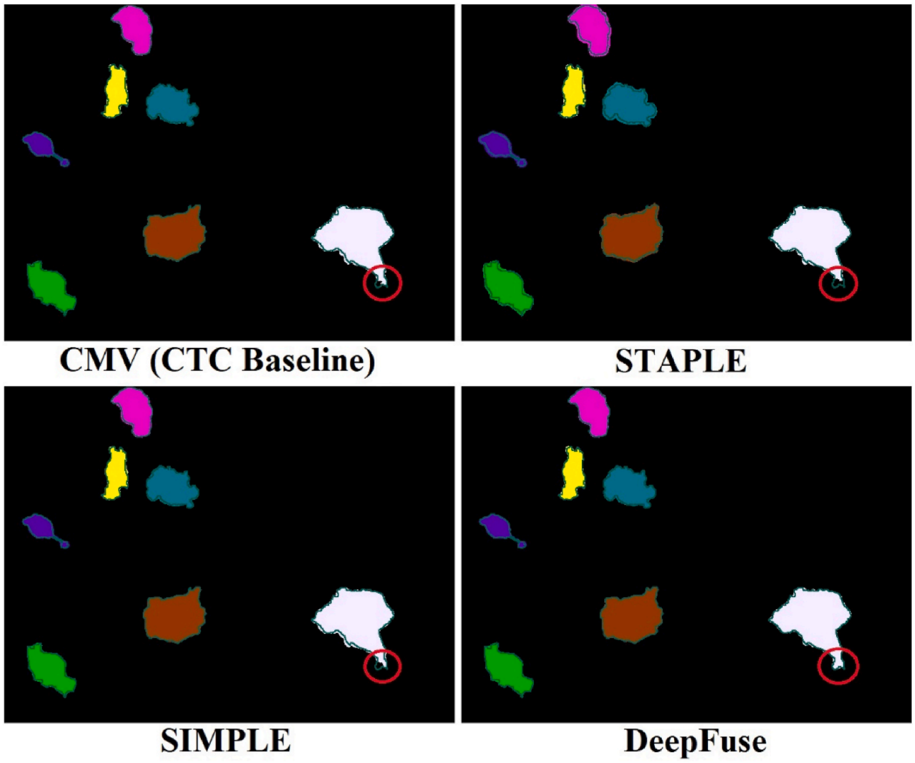

Fig. 3. Fusion results of the first sequence of PhC-C2DH-U373, time point 102, using the raters in Fig. 4. The dark green outline shows the boundaries of human-origin segmentation mask. All four fusion methods can handle the over-segmented masks, in different levels though, by shrinking the masks. However, when some of the raters (10 out of 15 in this particular example, please see Fig. 4) fail to segment a particular cell region partially or completely, only the DeepFuse method can recover such regions (please see the circled area). In other words, among these four fusion methods, DeepFuse is the only method that can enlarge an under-segmented instance when the majority of inputs are under-segmented. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

3.5. SIMPLE

SIMPLE is a weighted majority voting-based label fusion algorithm [5] which iteratively internally re-weights the input segments to remove quality-wise outlying inputs and fuses the rest of them. It has been successfully applied in medical imaging field so far. In the cell imaging domain, SIMPLE, in its available original implementation, cannot be straight-forwardly applied because it fuses all existing labels in a given source image at once, as if all should represent the same object. This property of the algorithm causes merging of the closely located masks during the fusion process. Hence, some of the labels are lost during fusion. To overcome this issue, we split the input images into single-mask images so that each image contains at most one label. Although this modification makes the algorithm slower, it guarantees to preserve the existing labels in the outputs.

3.6. STAPLE

STAPLE is a statistical label fusion algorithm that has been predominantly applied in the medical domain [8]. It produces silver-standard masks using an iterative process based on expectation maximization to incorporate annotator assessment. We have used the original implementation of STAPLE to produce the output which is actually a probability map that we afterwards thresholded by $p = 0.5$, as it is commonly done in its applications. As with DeepFuse and SIMPLE, we split the input images into single-mask images. Therefore, inputs of the three tested fusion methods are of the same format.

3.7. Experimental results

The DeepFuse approach produced more accurate segmentation masks than relevant approaches in 80 % of the 26 tested videos. Except for the *C. Elegans* training video, DeepFuse performed better than any of its input raters. This is, nevertheless, the case also for the other fusion approaches. Often, and that was also the case for the CTC baseline, the silver-standard reference datasets achieved accuracy better than declared human accuracy for the sequence in question. The quantitative results are presented in Table 3. Selected qualitative results are presented in Figs. 3 and 4. These figures illustrate the fact that, unlike STAPLE, SIMPLE, and CTC Baseline; DeepFuse does not always need the majority of votes to segment an image element thanks to the learned filters, which provide a specialized segmentation correction approach for each video sequence for which a different model is trained. This added capability helped DeepFuse outperform the compared methods in most of the video sequences analyzed in this paper. Thus, the fused segmentation masks created by DeepFuse may offer lower curation time for human experts.

In almost all cases in Table 3, DeepFuse outperformed the SIMPLE and STAPLE fusion algorithms. In 21 out of 26 sequences, DeepFuse was the best fusion algorithm, and in the remaining five cases its performance was very close to the best fusion algorithm. SIMPLE and STAPLE, just like any pure pixel-wise voting-based fusion approach, suffered from under-segmentation problem when a significant portion of the inputs was under-segmented. This is where DeepFuse shows its difference; DeepFuse can handle under-segmented input masks by filling holes or enlarging a mask (Figs. 3 and 4) thanks to its specialized convolutional image filters for each rater. To emphasize the importance of this new skill acquired by DeepFuse; the method does not follow the majority when they are wrong at some specific pixels, it first corrects the raters at those pixels and then fuses. This property introduces a new layer of quality improvement mechanism compared to the traditional fusion methods, and makes DeepFuse unique.

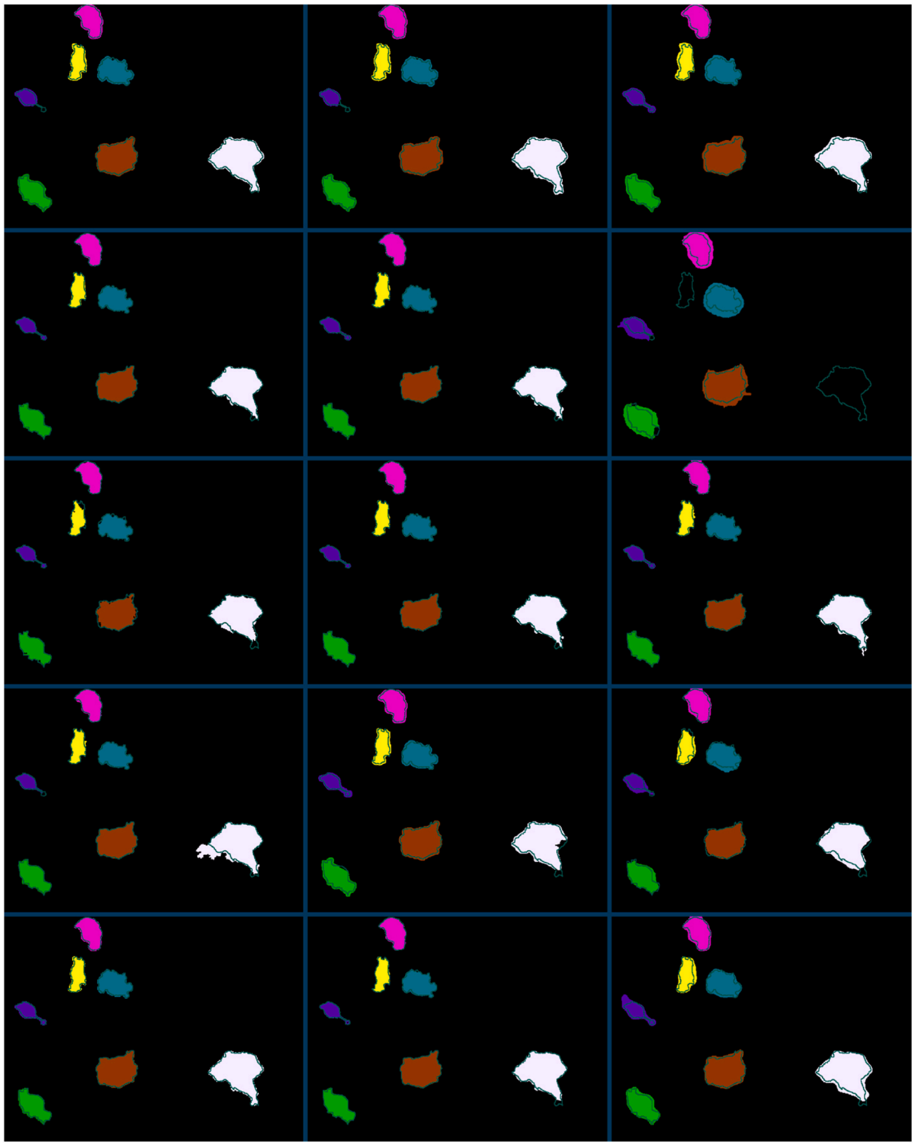

Fig. 4. All raters (pre-processed) from the first sequence of PhC-C2DH-U373, time point 102. Dark green outlines around the cell masks indicate the instance boundaries from the gold-standard annotations. Please note that the protrusion of the white cell instance (bottom-right) is not segmented well by some of the raters. Also notice that the number of masks per input may vary because, as the example shows here, some raters may fail to detect some cells. In such cases, we do not eliminate the particular rater and feed an empty segmentation mask to the network for the corresponding object. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

The convolutional stage can also be considered as a post-processing stage for the rater results (see Fig. 1, the part to the left of the yellow arrows) because input segmentation masks are refined using know-how learned from gold-standard annotations, allowing for a noticeable increase in the fusion quality (as exemplified in Figs. 3 and 4). In this example case, only four out of 15 raters were able to segment the bottom protrusion of the gray cell completely. It is clearly visible that CTC Baseline, STAPLE and SIMPLE failed to recover/segment such areas completely. In this particular example, DeepFuse was able to recover the whole protrusion area thanks to its convolutional post-processing (refinement) stage. In order to study the effect of the input quality threshold on the fusion results, we also conducted a study with varying number of inputs while gradually increasing the threshold for input rater results. Experimental results (see Table 3) show that the threshold values that achieved best fusion results with DeepFuse vary between 44.6 % and 95.8 %. While SIMPLE and STAPLE require top-1 or top-3 inputs for the most of the sequences analyzed, DeepFuse usually requires top-5 inputs or even more. This observation is related to the supervised learning mechanism of DeepFuse; more diverse training samples often increase the performance of trained models. Hence, in our experiments higher number of raters (i.e., lower input quality threshold) are usually favored for DeepFuse compared to traditional methods like SIMPLE and STAPLE. It should be noted that for seven out of 26 analyzed sequences, DeepFuse achieved the best results with input quality thresholds smaller than 50 %.

Our experimental results support the findings of [16,17] (See Section 1); DeepFuse could create a better rater than all available input raters for all 26 tested sequences. It is also observed that DeepFuse can produce better results than the CTC Baseline silver corpora in most of the CTC training datasets. Hence, DeepFuse will be a strong candidate for the creation of future versions of the CTC Silver Truth corpora to increase the quality of computer-origin segmentation annotations.

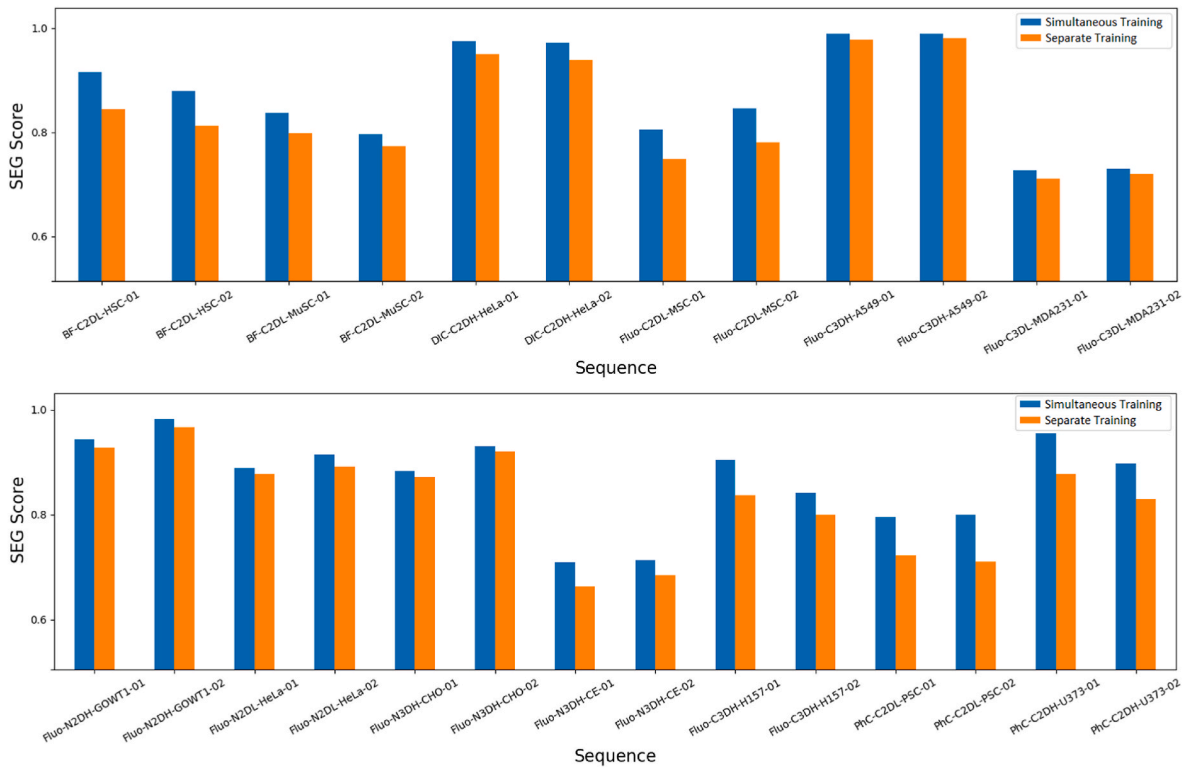

Fig. 5. The comparison between simultaneous training (DeepFuse) and separate training (ablated alternation of DeepFuse). In the separate training case, N separate networks are trained separately and then the results are fused with pixel-wise summation and sigmoid activation. Thus, these separate networks tend to prioritize improving the quality of individual rater results rather than the fusion result. However, in the simultaneous training case, which is the default DeepFuse setting, there is only one network for all N raters, which means that the objective of the loss function is to produce the best-possible fusion result. This advantage of the simultaneous training is clearly apparent in the experimental results.

3.8. Ablation studies

The DeepFuse architecture, which is depicted in Fig. 1, is training refining layers for each rater simultaneously with a single loss function. In a different scenario, we also trained N separate networks, and then applied concatenation and sigmoid activation to the separately trained segmentation masks to create fused masks (see Fig. 5). In this comparison, we observed that simultaneous training of all N raters' segmentation masks outperformed the separate training approach in all training sequences without exceptions. This difference can be explained by the fact that simultaneous training includes all raters' segmentation masks in a single loss function which is more in-line with the purpose of finding a consensus of them. Separately trained networks are driven by N individual loss functions which prioritize improving the individual results of raters rather than the fused masks. In other words, simultaneous training (Fig. 1) aims for the best-possible consensus result (fused mask), whereas separate training of inputs aims to bring out the best-possible versions of individual inputs, which may cause unintentional quality degradation in the consensus results. Separate training of rater results can be helpful when one wants to refine the results of a single rater. For multi-rater cases, our data clearly show superiority of the simultaneous training over separate training on the results of all raters.

4. Discussion

The experimental results indicate that DeepFuse outperforms traditional fusion methods like SIMPLE and STAPLE in 80 % of the tested cases, showcasing its robustness in producing accurate and reliable segmentation masks. This superior performance can be attributed to DeepFuse's ability to apply specialized post-processing to each rater's output, effectively correcting under-segmentation issues that are common in traditional methods. The full-resolution architecture plays a crucial role in this, as it allows the network to maintain image detail and refine segmentation outputs without losing critical information. This capability is particularly advantageous when dealing with complex cell structures or subtle features that are often missed by simpler fusion algorithms. However, in the remaining 20 % of cases where DeepFuse did not outperform the other methods, the challenges can be linked to the inherent complexity of the datasets. For example, 3D images usually have more intriguing voxel dependencies, thus refinement is harder to apply. Furthermore, the volume enables for a larger variety of spatial patterns that needs to be present in the training set in sufficient amounts to allow the network to learn how to fix and fuse them. This implies that, in fact, more 3D results from more raters and less sparse 3D ground-truth is required compared to 2D setting, when in reality the opposite is typically true. Assuming then that 2D refinement and fusion is more likely to be trained better, DeepFuse is approaching also 3D images in a slice-by-slice fashion. Even despite that, in some z-slices of 3D images, incomplete or unusual cell shapes may occur due to cross-sectioning. In the same line, another reason for the sub-optimal performance of DeepFuse is a limited representation of raters' behavior in training datasets. For example, certain datasets may contain highly heterogeneous cell populations or imaging conditions that pose difficulties for the network to generalize effectively, leading to performance gaps. Moreover, raters' behavior may not be represented completely in training datasets that we analyzed. As a common practice, larger annotated training datasets with increased sample variety may offer improved performance since DeepFuse is a fully supervised method. Our results suggest that while DeepFuse represents a significant advancement in label fusion, especially by introducing the full-resolution CNN architecture, there is still room for improvement; particularly in enhancing the network's ability to handle extremely challenging segmentation tasks or datasets with high variability.

Our experiments with three tested fusion methods feature M-best raters to study the impact of the rater quality threshold on the fusion results. Although one might expect a higher quality threshold would yield a higher fusion quality, the experimental results show that there is no clear pattern for the optimal quality threshold levels; they varied between 44.6 % and 95.8 % for DeepFuse (Table 3). In other words, the DeepFuse models trained on sets that also include raters of relatively lower quality were often able to produce more accurate fusion than models based solely on good raters. This observation can be explained by the supervised learning mechanism of CNNs. The training set that contains the closest representation in terms of true positive pixels compared to the human-annotator behavior achieves, naturally, the highest performance for the trained models. And having more false positive pixels in the relatively lower quality rater results would negatively impact the results, but this impact is minimized during the training via the loss function. The network could learn how to keep and even yield true positive pixels and to suppress false positive pixels even from the lower performing inputs. This behavior is, nevertheless, true also for the well performing inputs. This is opening new research directions towards understanding of why having only the well performing inputs is often not enough, how the lower performing inputs contribute to yield better fusion overall, and why the network seemed to learn the corrections better when lower performing inputs were involved in the training.

5. Conclusions

This study introduces DeepFuse, a novel full-resolution CNN-based multi-rater fusion network for biomedical image segmentation that advances the state of the art in annotation technologies. DeepFuse represents the first full-resolution CNN-based network designed to fuse multi-rater segmentation masks. Unlike traditional methods such as STAPLE and SIMPLE, our approach combines label fusion and mask refinement within a single CNN architecture, enabling more sophisticated and reliable annotation processes. The network's distinctive capabilities include: independent refinement of each rater's segmentation output, preservation of full image resolution, no prior assumptions about inter-pixel dependencies or object geometry, and adaptability across diverse imaging domains. Our experimental results demonstrate DeepFuse's superior performance, as evidenced by generating high-quality silver-standard segmentation masks, outperforming existing commonly used fusion methods and significantly reducing the quality gap between silver-standard and gold-standard annotations. The full-resolution architecture offers critical advantages in biomedical imaging, where preserving fine spatial details is diagnostically crucial, small structural features can significantly impact interpretation, and generalizability across different imaging datasets is essential to create robust annotations.

We anticipate that DeepFuse will influence future multi-rater fusion methods by demonstrating the importance of maintaining high-resolution in image processing, providing a framework for more efficient annotation workflows by employing fully-automatic mask refinement, and reducing the manual effort required in creating gold-standard annotations. The proposed approach not only advances technical capabilities in image segmentation but also offers a more time-efficient solution for researchers and medical professionals working with complex imaging datasets, enabling a faster pace in segmentation-related biomedical tasks.

CRediT authorship contribution statement

Cem Emre Akbaş: Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Vladimír Ulman:** Writing – review & editing, Software, Methodology, Investigation, Data curation, Conceptualization. **Martin Maška:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Michal Kozubek:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Availability of data and materials

Codes of the proposed DeepFuse architecture are available at http://github.com/cememreakbas/DeepFuse. Original datasets are freely available for download on Cell Tracking Challenge website at http://celltrackingchallenge.net/2d-datasets/ and http://celltrackingchallenge.net/3d-datasets/. Processed image data are available on reasonable request.

Ethics statement

This research did not require IRB approval because it is based on publicly available datasets.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Michal Kozubek, Vladimír Ulman and Cem Emre Akbas report financial support was provided by Ministry of Education, Youth and Sports of the Czech Republic. Martin Maška reports financial support was provided by Czech Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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